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Completely monotone families of solutions of n-th order linear differential equations and infinitely divisible distributions

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Completely Monotone Families of Solutions of n-th Order Linear Differential Equations and Infinitely Divisible Distributions (*).

PHILIP HARTMAN (**)

dedicated to Hans Lewy

1. – Introduction.

It was shown in [9] that if $x = I_\mu(t)$ is the unique solution of the modified Bessel differential equation

$$t^2x'' + tx' - (t^2 + \mu^2)x = 0 \tag{1.1}$$

satisfying $x \sim (t/2)^\mu/\Gamma(1+\mu)$ as $t \to 0$, then $I_\mu(t)$ is a completely monotone function of $\lambda = \mu^2 \geqq 0$ (for fixed $t > 0$); for the definitions of the solutions $I_\mu(t)$, $K_\mu(t)$ of (1.1), see [15], pp. 77-80. Thus, by the theorem of Hausdorff-Bernstein (cf. [16], p. 160), there exists a (unique) distribution function $W(r) = W(r, t, \infty)$ on $r \geqq 0$ satisfying

$$I_\mu(t) = I_0(t)\int_0^\infty \exp(-r\mu^2)\, W(dr, t, \infty) \qquad \text{for } \mu \geqq 0\,,$$

so that $W(0) = 0$, $W(\infty) = 1$, $W(r-0) = W(r)$, and $W(dr) \geqq 0$. The results below will imply the following generalization:

THEOREM 1.0. (*a*) $0 < t < \tau \leqq \infty$. *Then* $I_\mu(t)/I_\mu(\tau)$ *is a completely monotone function of* $\lambda = \mu^2 \geqq 0$, *so that there exists a* (*unique*) *distribution func-*

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tion $W(r) = W(r, t, \tau)$ *on* $r \geqq 0$ *satisfying*

$$I_\mu(t)/I_\mu(\tau) = [I_0(t)/I_0(\tau)]\int_0^\infty \exp(-r\mu^2)\, W(dr, t, \tau) \quad \text{for } \mu \geqq 0\,;$$

the distribution function

$$W(r) = W(r, t, \tau) \quad \text{is infinitely divisible} \tag{1.2}$$

for $0 < t \leqq \tau \leqq \infty$;

$$W(\cdot, \tau_1, \tau_n) = W(\cdot, \tau_1, \tau_2) * W(\cdot, \tau_2, \tau_3) * \ldots * W(\cdot, \tau_{n-1}, \tau_n) \quad \text{for } \tau_1 < \ldots < \tau_n \tag{1.3}$$

with $\tau_1 > 0$, $\tau_n \leqq \infty$;

(1.4) $W(r, t, \tau)$ is non-decreasing in r and t, and non-increasing in τ;

$$W(r, t, \tau) \to 0 \quad \text{as } t \to +0 \tag{1.5}$$

for $0 < \tau \leqq \infty$;

$$W(r, t, \tau) \to W(r, t, \infty) \quad \text{as } \tau \to \infty \text{ for } r \geqq 0\,, \tag{1.6}$$

and $t > 0$; *and*

$$W(r, t, \tau) \to \delta_0(r) \quad \text{as } t \uparrow \sigma,\ \tau \downarrow \sigma \tag{1.7}$$

with $0 < \sigma < \infty$,

$$W(r, t, \tau) \to \delta_0(r) \quad \text{as } \tau > t \to \infty\,, \tag{1.8}$$

where $\delta_0(r)$ *is the unit distribution function* (*i.e.*, $\delta(0)=0$ *and* $\delta(r)=1$ *for* $r>0$).

(*b*) *Also*, $1/K_\mu(t)$ *and* $K_\mu(\tau)/K_\mu(t)$ *are completely monotone functions of* $\lambda = \mu^2 \geqq 0$ *for* $0 < t < \tau < \infty$, *so that there exists a distribution function* $W(r) = W(r, t, \tau)$ *on* $r \geqslant 0$ *satisfying*

$$1/K_\mu(t) = [1/K_0(t)]\int_0^\infty \exp(-r\mu^2)\, W(dr, t, \infty) \quad \text{for } \mu \geqslant 0\,,$$

$$K_\mu(\tau)/K_\mu(t) = [K_0(\tau)/K_0(t)]\int_0^\infty \exp(-r\mu^2)\, W(dr, t, \tau) \quad \text{for } \mu \geqq 0\,.$$

The distribution function $W(r) = W(r, t, \tau)$ *satisfies* (1.2), (1.3) *with* $\tau_1 > 0$, $\tau_n \leqslant \infty$; (1.4), (1.5) *for* $0 < \tau \leqq \infty$; (1.6); (1.7) *with* $0 < \sigma < \infty$; *and* (1.8).

The arguments of [9] can be used to show that if the definition of $W(r, t, \tau)$ for $r \geqq 0$, $t > 0$ is extended by putting $W(r, t, \tau) = 0$ for $r < 0$ and/or $t = 0$, then $W(r, t, \tau)$ is continuous for $-\infty < r < \infty$ and $0 \leqq t < \tau (\leqq \infty)$, of class C^∞ for $-\infty < r < \infty$, $0 < t < \tau (\leqq \infty)$, and satisfies the parabolic equation

$$t^2 W_{tt} + t W_t - t^2 W = W_r ,$$

for fixed τ, on $-\infty < r < \infty$ and $0 < t < \tau (\leqq \infty)$.

The result of [9] concerning the complete monotony of $I_\mu(t)$ as a function of $\lambda = \mu^2 \geqq 0$ (for fixed $t > 0$) suggests the questions as to when a family of solutions $x = X(t, \lambda)$ of a 1-parameter family of differential equations

$$(1.9) \qquad D^n x + q_{n-1}(t) D^{n-1} x + \ldots + q_1(t) Dx - q(t, \lambda) x = 0 , \qquad D = d/dt ,$$

$(t, \lambda) \in T \times \Lambda$, is a completely monotone function of $\lambda \in \Lambda$ (for fixed $t \in T$) or when such a completely monotone family of solutions $x = X(t, \lambda)$ exists, and when the analogue of Theorem 1.0 is valid.

NOTATION. – Unless otherwise specified, an « interval » can be bounded or unbounded, and closed or open or neither. T is a t-interval with endpoints α and β, $-\infty \leqq \alpha < \beta \leqq \infty$ and $T^0 = (\alpha, \beta)$ is its interior. Λ is a λ-interval and Λ^0 its interior.

We shall make some of the following assumptions from time to time.

(A1) $q_1(t), \ldots, q_{n-1}(t) \in C^0(T)$ *and* $q(t, \lambda) \in C^0(T \times \Lambda)$. *For fixed* $\lambda \in \Lambda$, (1.9) *is disconjugate on* T; *i.e., no solution* $x(t) \not\equiv 0$ *has more than* $n-1$ *zeros on* T.

(A2$_k$) $\partial^m q / \partial \lambda^m$ *exists, is continuous, and satisfies* $(-1)^{m+k} \partial^m q / \partial \lambda^m \geqq 0$ *on* $T \times \Lambda^0$ *for* $m = 1, 2, \ldots$ *and a fixed* k (*in particular,* $(-1)^{k+1} \partial q(t, \cdot) / \partial \lambda$ *is completely monotone on* Λ^0 *for fixed* $t \in T$).

It is immaterial in assumption (A1) whether or not zeros are counted with multiplicities. For a more general result, see [6]; and for a simple proof for the linear case at hand, see [13].

When (A1) holds, (1.9) has first, second, ..., n-th principal solutions at $t = \beta$, say $x = \xi_1(t, \lambda), \ldots, \xi_n(t, \lambda)$; i.e., solutions satisfying $\xi_j(t, \lambda) \neq 0$ for t near β and $\xi_j(t, \lambda) / \xi_{j+1}(t, \lambda) \to 0$ as $t \to \beta$ for $j = 1, \ldots, n$; [7], pp. 328-347 or pp. 353-355 or [11]. (In [7], pp. 353-355, the assumption of « disconjugacy » is relaxed to « non-oscillatory at $t = \beta$ »). Furthermore, if $\tau \in T^0$,

then there exist unique principal solutions $\eta_1(t, \tau, \lambda), \ldots, \eta_n(t, \tau, \lambda)$ at $t = \beta$ such that $x = \eta_j(t, \tau, \lambda)$ satisfies

$$(1.10) \qquad D^{i-1}x = 0 \quad \text{for } 1 \leqq i < j \quad \text{and} \quad D^{j-1}x = 1 \quad \text{at } t = \tau\,;$$

[7], Theorem 7.2_n(vii), p. 332. The solution $x = \eta_j(t, \tau, \lambda)$ is called the j-th *special principal solution* at $t = \beta$, determined by τ. A first principal solution $x = \xi_1(t, \lambda)$ is unique up to a non-zero constant factor and $\xi_1(t, \lambda) \neq 0$ for $t \in T^0$ ([7], Theorem 7.1_n(i), pp. 331-332), so that $\eta_1(t, \tau, \lambda) = \xi_1(t, \lambda)/\xi_1(\tau, \lambda)$. Sometimes it will be convenient to assume

($A3_\sigma$) *Let $\sigma = \alpha$ or $\sigma = \beta$. There exists a family of non-negative first principal solutions $x = \xi_1(t, \lambda)$ at $t = \beta$ satisfying*

$$(1.11_\sigma) \qquad \xi_1(t, \lambda)/\xi_1(t, \mu) \to 1 \quad \text{as } t \to \sigma \text{ for } \lambda, \mu \in \Lambda\,.$$

This assumption is equivalent to the requirement that

$$(1.12) \qquad h(\lambda, \mu) \equiv \lim_{t \to \sigma} x_1(t, \lambda)/x_1(t, \mu) \text{ exists and } \neq 0 \text{ for } \lambda, \mu \in \Lambda$$

holds for arbitrary first principal solutions $x_1(t, \lambda) \geqq 0$ of (1.9). For in this case, $\xi_1(t, \lambda) = x_1(t, \lambda)/h(\lambda, \nu)$, for a fixed $\nu \in \Lambda$, satisfies (1.11_σ). The asymptotic integration theory of (1.9) gives simple sufficient conditions for the validity of (1.12), hence of (A3). Also, if $T = (\alpha, \beta]$ is closed and bounded on the right and $x = \xi_1(t, \lambda)$ is the solution of (1.9) satisfying the initial conditions (1.10) with $j = n$ and $\tau = \beta$, then (1.11_β) holds; contrast part (*b*) of the following theorem with the result in the Appendix 2 below.

THEOREM 1.1. *Assume* (A1) *and* ($A2_k$) *for a fixed k, $0 < k < n$.* (*a*) *Let $\tau \in T^0$ and $x = \eta_{n-k}(t, \tau, \lambda)$ be the special $(n-k)$-th principal solution at $t = \beta$ determined by τ. Then $D^i\eta_{n-k}(t, \tau, \lambda) \in C^0(T \times T^0 \times \Lambda^0)$ for $i = 0, \ldots, n$, and $\eta_{n-k}(t, \tau, \cdot)$ is completely monotone on Λ^0, for fixed (t, τ) with $\alpha < \tau \leqq t < \beta$.* (*b*) *If, in addition, $n - k = 1$, $\Lambda^0 \supset [0, \infty)$, and $x = \xi_1(t, \lambda)$ is a first principal solution, then, for $\alpha < t \leqq \tau < \beta$, there exists a distribution function $W(r) = W(r, t, \tau)$ on $r \geqq 0$ such that*

$$(1.13) \qquad \xi_1(\tau, \lambda)/\xi_1(t, \lambda) = [\xi_1(\tau, 0)/\xi_1(t, 0)]\int_0^\infty \exp(-\lambda r)\, W(dr, t, \tau) \quad \text{for } \lambda \geqq 0\,.$$

The distribution function $W(r) = W(r, t, \tau)$ satisfies (1.2), (1.3) *with $\tau_1 > \alpha$ and $\tau_n < \beta$;* (1.4); *and* (1.7) *with $\alpha < \sigma < \beta$.*

REMARK 1. In Theorem 1.1(b), condition « $\Lambda^0 \supset [0, \infty)$ » can be replaced by « $\Lambda = (0, \infty)$ » provided that $\xi_1(t, \lambda)$, defined on $T \times \Lambda = T \times (0, \infty)$, has a continuous extension to $T \times [0, \infty)$; see [7], p. 335, or [5], p. 360, Corollary 6.6 and the comment following it. A similar remark applies to Theorem 1.2(b) and the last part of Lemma 2.2 below.

REMARK 2. If (A2$_k$) holds, then (A2$_{n-k}$) holds after the change of variables $t \to -t$. Thus, part (a) implies that if $x = \zeta_k(t, \tau, \lambda)$ is the k-th special principal solution of (1.9) at $t = \alpha$ determined by τ, i.e., $x(t) = \zeta_k(t, \tau, \lambda)$ satisfies the conditions $D^{i-1}x = 0$ for $1 \leqq i < k$ and $D^{k-1}x = (-1)^{k-1}$ at $t = \tau$, then $D^i\zeta_k(t, \tau, \lambda) \in C^0(T \times T \times \Lambda^0)$ for $i = 0, \dots, n$, and $\zeta_k(t, \tau, \cdot)$ is completely monotone on Λ^0, for fixed (t, τ) with $\alpha < t \leqq \tau < \beta$.

THEOREM 1.2. *Assume* (A1), (A2$_{n-1}$) *and* (A3$_\sigma$) *with* $\sigma = \alpha$ [*or* $\sigma = \beta$]. (a) *Then* $D^i\xi_1(t, \lambda) \in C^0(T \times \Lambda^0)$ *for* $0 \leqq i \leqq n$, *and* $\xi_1(t, \cdot)$ [*or* $1/\xi_1(t, \cdot)$] *is completely monotone on* Λ^0 *for fixed* $t \in T^0$. (b) *If, in addition,* $\Lambda^0 \supset [0, \infty)$, *then there exists a distribution function* $W(r) = W(r, \alpha, t)$ [*or* $W(r) = W(r, \tau, \beta)$] *on* $r \geqq 0$ *such that*

$$\xi_1(\tau, \lambda) = \xi_1(\tau, 0)\int_0^\infty \exp(-\lambda r)\, W(dr, \alpha, \tau) \quad \text{for } \lambda \geqq 0 \tag{1.14}$$

and $\tau \in T^0$ [*or such that*

$$1/\xi_1(t, \lambda) = [1/\xi_1(t, 0)]\int_0^\infty \exp(-\lambda r)\, W(dr, t, \beta) \quad \text{for } \lambda \geqq 0 \tag{1.15}$$

and $t \in T_0$]; *also* $\tau_1 = \alpha$ [*or* $\tau_n = \beta$] *is admissible in* (1.3); $W(r, t, \tau) \to \delta_0(r)$ *as* $t < \tau \to \alpha$ [*or* $\tau > t \to \beta$]; *and* $W(\cdot, \alpha, \tau)$ [*or* $W(\cdot, t, \beta)$] *is infinitely divisible.*

REMARK 3. It will be clear from the proofs that Theorems 1.1 and 1.2 remain valid if (1.9) is replaced by an equation involving quasi-derivatives of the form

$$\left\{p_{M+1}^{-1} D p_M^{-1} D \dots p_1^{-1} D^K + \sum_{j=1}^{K-1} q_j D^j - q\right\} x = 0\,, \tag{1.9'}$$

the operators D^i for $i = 0, \dots, n$ are replaced by $L_i = D^i$ for $0 \leqq i < K$ and $L_{i+K} = p_{i+1} D \dots D p_1^{-1} D^K$ for $0 \leqq i \leqq M$, and assumption (A1) is replaced by

(A1′) $K \geqq 1$, $M \geqq 0$, *and* $n = K + M$; $0 < p_j = p_j(t) \in C^0(T)$ *for* $j = 1, \dots, M+1$; $q_j = q_j(t) \in C^0(T)$ *for* $j = 1, \dots, K-1$; $q = q(t, \lambda) \in C^0(T \times \Lambda)$; *and* (1.9′) *is disconjugate on* T *for fixed* $\lambda \in \Lambda$.

As an application, different from that concerning $I_\mu(t)$, we might mention that the results above imply that the Legendre differential equation

$$[(t^2-1)x']' - [\nu(\nu+1) + \mu^2/(t^2-1)]x = 0$$

has a family of principal solutions $x = X(t, \mu, \nu)$ at $t = 1$ on $T = (1, \infty)$ such that $X(t, \mu, \nu)$ is a completely monotone function of $\lambda = \mu^2 \geqq 0$ for fixed $t > 1$, $\nu \geqq 0$. In fact, from Theorem 1.2, we obtain $X(t, \mu, \nu)$ in the form $c(\mu, \nu) P_\nu^{-\mu}(t)$, where $c(\mu, \nu)$ can be given explicitly in terms of the Γ-function and $P_\nu^{-\mu}$ is a Legendre function of the first kind; cf. [4], p. 122, (3) and (5). Also, if $Q_\nu^\mu(t)$ is a Legendre function of the second kind, then $Q_\nu^\mu(t)$ is a principal solution at $t = \infty$ for fixed $\mu, \nu \geqq 0$ and, by Theorem 1.2, $\Gamma(\nu + \mu + 1) \cdot \cdot \exp(\pi i \mu)/Q_\nu^\mu(t)$ is a completely monotone function of $\lambda = \mu^2 > 0$ for fixed $t > 1$, $\nu \geqq 0$; cf. [4], p. 122, (5). We could also formulate an analogue of Theorem 1.0.

In Section 2, we state the main Lemma 2.1 concerning the complete monotony of certain Green's functions with respect to λ (for fixed s, t). Section 3 contains the proofs of Theorems 1.1 and 1.2. Section 4 gives somewhat different criteria for a family of solutions $X(t, \lambda)$ of second order differential equations to be completely monotone with respect to λ. Theorem 5.1 deals with a generalization of (1.1) in a neighborhood of a regular singular point. Finally, Section 6 contains interesting related observations about the Γ-function.

Appendix 1 concerns the situation when m is restricted to a finite range in condition (A2$_k$). Appendix 2 deals with the Cauchy functions of a 1-parameter family of n-th order linear differential equations. Appendix 3 extends from $j = 1$ to $1 \leqq j < n$ a known characterization of a j-th special principal solution of a disconjugate equation; cf. Theorem 7.1$_n$, [7], p. 330 for the case $j = 1$.

2. – On Green's functions.

Assume (A1) and fix $[a, b] \subset T$ and k, $0 < k < n$. Write the left side of (1.9) as $L(\lambda)x$, where $L(\lambda) = D^n + q_{n-1}D^{n-1} + \ldots + q_1 D - q$ is a differential operator. Consider the boundary value problem on $[a, b]$ consisting of the differential equation $L(\lambda)x = 0$ and boundary conditions

$$(D^{i-1}x)(a) = 0 \quad \text{for } i = 1, \ldots, n-k, \tag{2.1}$$

$$(D^{i-1}x)(b) = 0 \quad \text{for } i = 1, \ldots, k. \tag{2.2}$$

The fact that (1.9) is disconjugate on T, hence on $[a, b]$, implies the existence of a Green's function $G_{ab}(t, s, \lambda)$ defined on $[a, b]\times[a, b]\times\Lambda$ such that if $h(t)\in C^0[a, b]$ or $h(t)\in L^2(a, b)$, then

$$L(\lambda)x = h \tag{2.3}$$

has a unique solution satisfying (2.1), (2.2), and x is given by

$$x(t) = \int_a^b G_{ab}(t, s, \lambda)h(s)\,ds\,. \tag{2.4}$$

Recall that G_{ab} is uniquely determined by the conditions: (i) as a function of t, $x(t) = G_{ab}(t, s, \lambda)$ is a solution of (1.4) on $[a, s)$, $(s, b]$ and satisfies (2.1), (2.2); (ii) $D^{i-1}x(t) = D^{i-1}G_{ab}(t, s, \lambda)\in C^0([a, b]\times[a, b])$ for fixed $\lambda\in\Lambda$ and $i = 1, \ldots, n-1$, and $D^{n-1}G_{ab}(s+0, s, \lambda) - D^{n-1}G_{ab}(s-0, s, \lambda) = 1$.

The main lemma is the following

LEMMA 2.1. *Assume* (A1), (A2_k) *with fixed* k, $0<k<n$, *and* $[a, b]\subset T$. *Then* $(-1)^k G_{ab}(t, s, \cdot)$ *is completely monotone on* Λ *for fixed* s, $t\subset[a, b]$.

McKean [12] contains a similar result for self-adjoint (possibly singular) boundary value problems with $n = 2$, $q(t, \lambda) = \lambda + Q(t)\geqq\lambda\geqq 0$. We simplify his operator-theoretic proof.

REMARK. Lemma 2.1 implies that if $h(t) = h(t, \lambda)\in C^0([a, b]\times\Lambda)$ has the property that $h(t, \cdot)$ is completely monotone on Λ for fixed $t\in[a, b]$, then the unique solution $x(t) = x(t, \lambda)$ of (2.1)-(2.3) given by (2.4) is completely monotone on Λ for fixed $t\in[a, b]$.

The proofs of Lemma 2.1 and Theorem 1.1-1.2 can be modified to obtain the following:

LEMMA 2.2. *Let* $0<p(t)\in C^0(T)$, $q(t, \lambda)\in C^0(T\times\Lambda)$ *satisfy* (A2_1), *and*

$$(p(t)x')' - q(t, \lambda)x = 0 \tag{2.5}$$

disconjugate on T *for fixed* λ. *Let* $x = Y(t, \lambda)$, $Z(t, \lambda)$ *be non-negative (first) principal solutions of* (2.5) *at* $t = \alpha, \beta$, *and suppose that* Y, Z *are linearly independent and normalized by* $p(t)[ZY' - Z'Y] = 1$. *Then* $G(t, s, \lambda)$, *defined as* $Y(t, \lambda)Z(s, \lambda)$ *or* $Y(s, \lambda)Z(t, \lambda)$ *according as* $t\leqq s$ *or* $t\geqq s$, *is a completely monotone function of* $\lambda\in\Lambda^0$ *for fixed* $s, t\in T$. *If either* $s\leqq\tau\leqq t$ *or* $t\leqq\tau\leqq s$,

then $G(t, s, \cdot)/G(\tau, \tau, \cdot)$ *is completely monotone on* Λ^0. *If also* $\Lambda^0 \supset [0, \infty)$, *then there exists a distribution function* $P(r) = P(r, t, s, \tau)$ *on* $r \geqq 0$ *satisfying*

$$G(t, s, \lambda)/G(\tau, \tau, \lambda) = [G(t, s, 0)/G(\tau, \tau, 0)] \int_0^\infty \exp(-r\lambda) P(dr, t, s, \tau)$$

for $\lambda \geqq 0$, *and* $P(\cdot, t, s, \tau)$ *is infinitely divisible.* Note that

$$G(t, s, \lambda)/G(\tau, \tau, \lambda) = [Y(s, \lambda)/Y(\tau, \lambda)] \cdot [Z(t, \lambda)/Z(\tau, \lambda)] \quad \text{if } s \leqq \tau \leqq t .$$

PROOF OF LEMMA 2.1. It will be clear from the explicit construction of $G_{ab}(t, s, \lambda)$ in the proof of Theorem 1.1 that $\partial^{i+j+m} G_{ab}(t, s, \lambda)/\partial t^i \partial s^j \partial \lambda^m$ exists and is continuous on $[a, b] \times [a, b] \times \Lambda$[or Λ^0] if $m = 0$ [or $m > 0$] and either $0 \leqq i, j \leqq n-2$ or $0 \leqq i, j \leqq n$ and $t \neq s$. It is known ([1], [10], [14]; cf. [2], p. 105) that $(-1)^k G_{ab}(t, s, \lambda) \geqq 0$ on $[a, b] \times [a, b] \times \Lambda$.

If $t \neq s$, then $L(\lambda) G(\cdot, s, \lambda) = 0$. Differentiation of this equation with respect to $\lambda \in \Lambda^0$ gives (2.3), where $x = \partial G_{ab}(\cdot, s, \lambda)/\partial \lambda$ and $h = G_{ab}(\cdot, s, \lambda) \cdot \partial q(\cdot, \lambda)/\partial \lambda$. As functions of t, h is continuous and $x \in C^n[a, b]$ satisfies (2.1)-(2.2). Hence (2.4) holds, i.e.,

$$(2.6) \qquad (-1)^k \partial G_{ab}(t, s, \lambda)/\partial \lambda = \\ = \int_a^b [(-1)^k G_{ab}(t, r, \lambda)] \cdot [(-1)^k q_\lambda(r, \lambda)] \cdot [(-1)^k G_{ab}(r, s, \lambda)] \, dr \leqq 0 .$$

Successive differentiations with respect to $\lambda \in \Lambda_0$ and an obvious induction give Lemma 2.1.

3. – Proof of Theorem 1.1(a).

Let $x(t) = X(t, \tau, \lambda)$ be the solution of (1.4) satisfying the initial conditions (1.10) with $j = n$, so that $X(t, \tau, \lambda)$ is the Cauchy function for (1.9) for fixed $\lambda \in \Lambda$. It is clear that (A1) implies that $\partial^i X/\partial t^j \partial \tau^k \in C^0(T \times T \times \Lambda)$ for $j + k = i = 0, \ldots, n$. Also if (A2$_k$) holds, then $\partial^{i+j+m} X/\partial t^i \partial \tau^j \partial \lambda^m \in C^0(T \times T \times \Lambda^0)$ for $i, j = 0, \ldots, n$ and arbitrary m.

Let $x(t) = Y(t, a, s, b, \lambda)$, $Z(t, a, s, b, \lambda)$ be the solutions of (1.9) such that

$$(3.1) \qquad G_{ab}(t, s, \lambda) = Y(t, a, s, b, \lambda) \quad \text{or } Z(t, a, s, b, \lambda)$$

according as $a \leqq t \leqq s$ or $s \leqq t \leqq b$. Thus, condition (ii), following (2.4) and

defining G_{ab}, implies that $Z - Y = X(t, s, \lambda)$, while (i) implies that $x(t) = Y$ satisfies (2.1) and $x(t) = Z$ satisfies (2.2). Since $Y = Z - X$, $x = Z$ is determined by the boundary conditions

$$(3.2) \qquad [D^{i-1}Z]_{t=a} = [D^{i-1}X]_{t=a} \qquad 1 \leqq i \leqq n-k\,,$$
$$[D^{i-1}Z]_{t=b} = 0 \qquad \text{if } 1 \leqq i \leqq k\,.$$

Similarly, $x = Y$ is determined by

$$(3.3) \qquad [D^{i-1}Y]_{t=a} = 0 \qquad \text{if } 1 \leqq i \leqq n-k\,,$$
$$[D^{i-1}Y]_{t=b} = -[D^{i-1}X]_{t=b} \qquad \text{if } 1 \leqq i \leqq k\,.$$

Let $x(t) = X^j(t, s, \tau, \lambda)$ be the unique solution of (1.9) satisfying the boundary conditions

$$(3.4) \qquad [D^{i-1}X^j]_{t=s} = 0 \qquad \text{if } 1 \leqq i \leqq n-j\,,$$
$$[D^{i-1}X^j]_{t=\tau} = 0 \qquad \text{or } 1 \text{ if } 1 \leqq i < j \text{ or } i = j\,.$$

Thus, by (3.2) and (3.3), we have

$$(3.5) \qquad Z(t, a, s, b, \lambda) = \sum_{j=1}^{n-k} X^j(t, b, a, \lambda) \cdot D^{j-1}X(a, s, \lambda)\,,$$

$$(3.6) \qquad Y(t, a, s, b, \lambda) = -\sum_{j=1}^{k} X^j(t, a, b, \lambda) \cdot D^{j-1}X(b, s, \lambda)\,.$$

This makes it clear that Y, Z have smoothness properties similar to those of X.

PROPOSITION 3.1. *Assume* (A1) *and* ($A2_k$) *with* k *fixed*, $0 < k < n$. *Then* $(-1)^{k+1}X^k(t, a, s, \cdot)$, *for* $a < t < s$, *and* $X^{n-k}(t, b, s, \cdot)$, *for* $s < t < b$, *are completely monotone on* Λ.

PROOF. The definition of the Cauchy function $X(t, \tau, \lambda)$ implies that $D^{j-1}X(t, \tau, \lambda) \sim (t-\tau)^{n-j}/(n-j)!$ as $t \to \tau$ uniformly on compacts of Λ. Hence, by (3.5) and (3.6),

$$X^k(t, a, s, \lambda) = -\lim_{b \to s} Y(t, a, s, b, \lambda)(n-k)!/(b-s)^{n-k}\,,$$
$$X^{n-k}(t, b, s, \lambda) = \lim_{a \to s} (-1)^k Z(t, a, s, b, \lambda) k!/(s-a)^k\,.$$

Thus, Proposition 3.1 follows from Lemma 2.1 and (3.1); cf. [16], pp. 147 ff., or Appendix 1 below.

PROPOSITION 3.2. *Assume* (A1) *and let* $\lambda \in \Lambda$ *be fixed. Then*

$$\eta_j(t, s, \lambda) = \lim_{b \to \beta} X^j(t, b, s, \lambda) \qquad \text{for } j = 1, \dots, n-1 . \tag{3.7}$$

This limit is valid in C^n *on arbitrary* (t, s)*-compacts of* $T \times T^0$.

REMARK. This proposition and (3.5) imply, for example, that

$$(-1)^k \sum_{j=1}^{n-k} \eta_j(t, s, \cdot) D^{j-1} X(a, s, \cdot)$$

is completely monotone on Λ^0 for fixed (a, s, t), $a < s < t$ and $a, t \in T$. Proposition 3.2 is contained in Appendix 3 below.

It is now easy to complete the proof of Theorem 1.1(a). Propositions 3.1, 3.2 imply that $\eta_{n-k}(t, s, \cdot)$ is completely monotone on Λ^0 for fixed (s, t), $\alpha < s \leqq t < \beta$. In particular, it is continuous and non-increasing with respect to λ for fixed (s, t). It therefore follows from Dini's theorem that $\eta_{n-k}(t, s, \lambda) \to \eta_{n-k}(t, s, \lambda_0)$, as $\lambda \to \lambda_0 \in \Lambda^0$, uniformly on (t, s)-compacts of $\alpha < s \leqq t < \beta$, hence of $T \times T^0$. Consequently, $\eta_{n-k}(t, s, \lambda) \in C^0(T \times T^0 \times \Lambda^0)$.

Hence, $D^i \eta_{n-k}(t, s, \lambda) \in C^0(T \times T^0 \times \Lambda^0)$ when $i = 0, \dots, n$, for a solution of (1.9) is uniquely determined by its values at n distinct t-values $\big($i.e., if $x_1(t, \lambda), \dots, x_n(t, \lambda)$ are linearly independent solutions of (1.9) and $t_1 < \dots < t_n$, then $\det(x_i(t_j, \lambda)) \neq 0\big)$. (For an analogous non-linear result, see the proof of Lemma I 1.1, [8], pp. 207-210. In the linear case at hand, a proof need not use the disconjugacy of (1.4), cf. Lemma 7.1, [5], p. 479.)

PROOF OF THEOREM 1.1(b). Since $\eta_1(t, t, \cdot) = \xi_1(\tau, \cdot)/\xi_1(t, \cdot)$ is completely monotone on $\Lambda = [0, \infty)$ for fixed (t, τ) with $\alpha < t \leqq \tau < \beta$ by part (a), the existence of a unique distribution function $W(r) = W(r, t, \tau)$ satisfying (1.13) follows from the Hausdorff-Bernstein theorem. The relation (1.3) for $\tau_1 < \dots < \tau_n$ with $\tau_1, \tau_n \in T$ is a consequence of (1.13) and the standard theorem on the products of Laplace-Steltjes transforms. The statement (1.4) follows from (1.3); in fact, if $\tau_1 < \sigma < \tau_2$, then (1.3) implies that

$$W(r, \tau_1, \tau_2) = \int_0^r W(r - \varrho, \tau_1, \sigma)\, dW(\varrho, \sigma, \tau_2) = \int_0^r W(r - \varrho, \sigma, \tau_2)\, dW(\varrho, \tau_1, \sigma)$$

so that

$$W(r, \tau_1, \tau_2) \leqq W(r, \tau_1, \sigma) \quad \text{and} \quad W(r, \tau_1, \tau_2) \leqq W(r, \sigma, \tau_2) ,$$

which implies (1.4). The statement (1.7) is clear from (1.13) since $\eta_1(\tau, t, \lambda) = \xi_1(\tau, \lambda)/\xi_1(t, \lambda) \to 1$ as $t, \tau \to \sigma$ uniformly on (σ, λ)-compacts of $T^0 \times [0, \infty)$. Furthermore, the monotony condition (1.4) and (1.7) imply, by virtue of Dini's theorem, that (1.7) holds uniformly on compact (r, σ)-subsets of $\{r > 0\} \times T^0$. The infinite divisibility of $W(\cdot, t, \tau)$, $\alpha < t \leqq \tau < \beta$, follows from (1.3) and the uniformity of (1.7); cf. [3], pp. 128-135.

PROOF OF THEOREM 1.2(a). By Theorem 1.1, $\eta_1(\tau, t, \cdot) = \xi_1(\tau, \cdot)/\xi_1(t, \cdot)$ is completely monotone on Λ^0 if $\alpha < t \leqq \tau < \beta$. Thus if (A3$_\alpha$) holds and $\mu \in \Lambda$ is fixed, $\xi_1(\tau, \cdot) = \lim \eta_1(\tau, t, \cdot)\xi_1(t, \mu)$, as $t \to \alpha$, is completely monotone on Λ^0 for fixed $t \in T^0$. Similarly, if (A3$_\beta$) holds, then $1/\xi_1(t, \cdot)$ is completely monotone on Λ^0 for fixed $t \in T^0$. The asserted smoothness properties of $\xi_1(t, \lambda)$ follow as in the proof of Theorem 1.1(a).

PROOF OF THEOREM 1.2(b). This is similar to the proof of Theorem 1.1(b) and will be omitted.

4. – Use of asymptotic estimates.

It was mentioned in [9], although no details were given, that a proof that $I_\mu(t)$ is a completely monotone function of $\lambda = \mu^2 \geqq 0$ could be based on a knowledge of the asymptotic behavior of $I_\mu(t)$ near $t = 0$ and $t = \infty$, instead of using Green's functions. A generalization of such a proof can be obtained from the following simple proposition.

Theorem 4.1. *In the differential equation*

$$(p(t)x')' - q(t, \lambda)x = 0\,, \tag{4.0}$$

let $0 < p(t) \in C^0(T)$, $q(t, \lambda)$ *satisfies assumption* (A2$_1$) *and* $q(t, \lambda) \geqq 0$. *Let* $x = X(t, \lambda)$ *be a solution of* (4.0) *such that* $X, X' \in C^0(T \times \Lambda)$, *and* $\partial^n X/\partial\lambda^n$ *exists and is continuous on* $T \times \Lambda^0$ *for* $n = 0, 1, \ldots$. *Then* $X(t, \cdot)$ *is completely monotone on* Λ (*for fixed* $t \in T$) *if and only if*

$$0 \leqq \limsup_{t \to \sigma} (-1)^n \partial^n X(t, \lambda)/\partial\lambda^n (\leqq \infty) \qquad \text{for } n = 0, 1, \ldots \tag{4.1$_\sigma$}$$

holds for $\sigma = \alpha, \beta$ *and all* $\lambda \in \Lambda^0$.

In contrast to Theorems 1.1 and 1.2, it is not supposed that $x = X(t, \lambda)$ is a principal solution. A variant of Theorem 4.1 is the following:

THEOREM 4.2. *Let $p(t)$, $q(t, \lambda)$ be as in Theorem 4.1 and $x = X(t, \lambda)$ a principal solution* (4.0) *at $t = \beta$ such that $\partial^n X/\partial\lambda^n$ exists and is continuous on $T \times \Lambda^0$ for $n = 0, 1, ..$ and that either*

$$(4.2) \qquad \int^{\beta} dt/p(t) = \infty \qquad \text{or} \qquad \lim_{t\to\beta} X(t, \lambda) = 0 \qquad \text{for } \lambda \in \Lambda .$$

Then $X(t, \cdot)$ is completely monotone on Λ if and only if

$$(4.3) \qquad 0 \leqq \limsup_{t\to\alpha} (-1)^n \partial^n X(t, \lambda)/\partial\lambda^n (\leqq \infty) \qquad \text{for } \lambda \in \Lambda^0 ,\ n = 0, 1, \dots .$$

PROOF OF THEOREM 4.1. Since the necessity of (4.1_α) and (4.1_β) for $\lambda \in \Lambda^0$ is clear, we only prove the sufficiency. Since $q(t, \lambda) \geqq 0$, a standard maximum principle shows that a solution of (4.0), for fixed λ, cannot have a negative minimum interior to T. Thus (4.1_σ) for $n = 0$ and $\sigma = \alpha, \beta$ imply the case $n = 0$ of

$$(4.4_n) \qquad (-1)^n X^{(n)} \geqq 0 \quad \text{on} \quad T \times \Lambda^0 , \qquad \text{where } X^{(n)} = \partial^n X/\partial\lambda^n .$$

Let $n - 1 \geqq 0$ and assume (4.4_k) for $k = 0, \dots, n-1$. If $x = X(t, \lambda)$ in (4.0), n differentiations with respect to λ on Λ^0 give

$$(4.5) \qquad (p(t)y')' - q(t, \lambda)y = -F_n(t, \lambda) ,$$

where

$$(4.6) \qquad y = (-1)^n X^{(n)} , \qquad F_n = \sum_{k=0}^{n-1} C_{nk}(-1)^{n-k+1} q^{(n-k)}(-1)^k X^{(k)} \geqq 0 ,$$

$q^{(m)} = \partial^m q/\partial\lambda^m$, and $C_{nk} = n!/k!(n-k)!$. Another appeal to the maximum principle shows that, for a fixed λ, y cannot have a negative minimum interior to T. Hence (4.1_σ) for $\sigma = \alpha, \beta$ implies (4.4_n). This completes the proof.

PROOF OF THEOREM 4.2. Since one-half of Theorem 1.2 is trivial, we prove only the sufficiency of (4.3) for the complete monotony of $X(t, \cdot)$. We first consider the case

$$(4.7) \qquad X(t, \lambda) \to 0 \quad \text{as } t \to \beta \text{ for } \lambda \in \Lambda$$

of (4.2). By induction, we shall prove (4.4_n) and

$$(4.8_n) \qquad X^{(n)}(t, \cdot) \to 0 \quad \text{as } t \to \beta \text{ in } L^1_{\text{loc}}(\Lambda^0) .$$

Of course, (4.8_n) is trivial if $T=(\alpha,\beta]$ is closed on the right, by virtue of (4.7) and the continuity of $X^{(n)}$ on $T\times\Lambda$.

The case $n=0$ of (4.4_n) follows as in Theorem 4.1. Also the case $n=0$ of (4.8_n) follows from (4.7) and Lebesgue's monotone convergence theorem. Assume (4.4_k) and (4.8_k) for $k=0,\dots,n-1$, and differentiate (4.0) with respect to $\lambda\in\Lambda^0$ to obtain (4.5), (4.6). Suppose, if possible, that

$$y(\tau,\lambda_0)<0 \quad \text{for some } (\tau,\lambda_0)\in T\times\Lambda^0\,. \tag{4.9}$$

Note that if $T=(\alpha,\beta]$ is closed on the right, then $\tau\neq\beta$ by (4.7). Since y cannot have a negative minimum at an interior point of T, it follows from (4.3) that $y'(t,\lambda_0)\leqq 0$ and $y(t,\lambda_0)\leqq y(\tau,\lambda_0)<0$ for $t\geqq\tau$. By continuity, $y(\tau,\lambda)<0$ and similarly $y(t,\lambda)\leqq y(\tau,\lambda)\leqq y(\tau,\lambda_0)/2<0$ for $t\geqq\tau$ and λ near λ_0.

If $\lambda>\mu$, then we have

$$\begin{aligned}(-1)^{n+1}\Big\{X(t,\lambda)-\sum_{k=0}^{n-1}X^{(k)}(t,\mu)(\lambda-\mu)^k/k!\Big\}&=\\ =-\int_\mu^\lambda(\lambda-\nu)^{n-1}y(t,\nu)\,d\nu/(n-1)!\,.&\end{aligned} \tag{4.10}$$

The right side is not less than $-(\lambda-\mu)^n y(\tau,\lambda_0)/2n!>0$ for λ, μ near λ_0 and $t\geqq\tau$. Integrate (4.10) with respect to μ over an interval $[\varrho,\lambda]$ to obtain

$$\begin{aligned}(-1)^{n+1}X(t,\lambda)(\lambda-\varrho)-\sum_{k=0}^{n-1}\int_\varrho^\lambda X^{(k)}(t,\mu)(\lambda-\mu)^k\,d\mu/k!&=\\ =-\int_\varrho^\lambda(\lambda-\nu)^{n-1}(\nu-\varrho)\,y(t,\nu)\,d\nu/(n-1)!\,,&\end{aligned} \tag{4.11}$$

which is not less than $-(\lambda-\varrho)^{n+1}y(\tau,\lambda_0)/2(n+1)!>0$. This contradicts (4.7) and (4.8_k) for $k=0,\dots,n-1$, and so (4.9) cannot hold. Thus $y=(-1)^nX^{(n)}\geqq 0$ on $T\times\Lambda^0$, i.e., (4.4_n) holds. Furthermore, (4.7), (4.8_k) for $k=0,\dots,n-1$, and (4.11) imply (4.8_n). This completes the induction and the proof of Theorem 1.2 in the case (4.7) of (4.2).

We now consider the case $\int^\beta dt/p(t)=\infty$ of (4.2). By induction, we prove (4.4_n) and

$$X^{(n)}(t,\cdot)\text{ is bounded } t\to\beta \text{ in } L^1_{\mathrm{loc}}(\Lambda^0)\,. \tag{4.12$_n$}$$

This proof is similar to that above if it is noted that (4.9) implies not only $y'(t, \lambda_0) \leqq 0$ for $t \leqq \tau$, but also that $y(t, \lambda_0) \to -\infty$ as $t \to \beta$. This follows from a convexity argument since (4.5) shows that $d^2y(t, \lambda_0)/ds^2 \leqq 0$ for $0 \leqq s < \infty$, where $ds/dt = 1/p(t)$, $s(\tau) = 0$, and $t = t(s)$ is the inverse of $s = s(t)$.

5. – A regular singular point.

The next result conerns the case of the family of differential equations

$$t^2x'' + r_0 tx' - \left[q_0(\lambda) + \sum_{n=1}^{\infty} q_n t^n\right] x = 0 . \tag{5.1}$$

THEOREM 5.1. *Let r_0 be arbitrary and $q_1, q_2, \ldots$ non-negative constants such that the power series in* (5.1) *is convergent for $|t| < \beta (\leqq \infty)$. Let $q_0(\lambda) \in C^\infty(\Lambda)$ satisfy*

$$q_0(\lambda) > 0 \quad \text{and } \partial q_0/\partial\lambda \text{ is completely monotone on } \Lambda . \tag{5.2}$$

The indicial polynomial $P(\nu) = P(\nu, \lambda)$,

$$P(\nu, \lambda) = \nu(\nu - 1) + r_0\nu - q_0(\lambda) \tag{5.3}$$

has the (unique) positive zero $\nu = \nu(\lambda) > 0$,

$$\nu(\lambda) = (1 - r_0)/2 + [(1 - r_0)^2/4 + q_0(\lambda)]^{\frac{1}{2}} , \tag{5.4}$$

and $\partial\nu/\partial\lambda$ is completely monotone on Λ. Let $c(\lambda) \neq 0$ be arbitrary. Then, for fixed λ, (5.1) *has a solution*

$$X(t, \lambda) = c(\lambda) t^{\nu(\lambda)} \sum_{n=0}^{\infty} x_n(\lambda) t^n \tag{5.5}$$

on $0 < t < \beta$ such that $x_0(\lambda) = 1$ and $x_n(\lambda)$ is completely monotone on Λ for $n = 1, 2, \ldots$. Hence, if $c(\lambda)t^{\nu(\lambda)}$ is completely monotone on Λ for $0 < t < \beta_0 (\leqq \beta)$, then the same is true of (5.5). (*This is the case if $c(\lambda) \equiv 1$ and $\beta_0 \leqq 1$.*)

PROOF. We shall make use of the following simple fact in this proof and in the next section.

PROPOSITIOH 5.1. *Let $g(\mu)$ be completely monotone for $\mu \geqq 0$. Let $\mu = \varphi(\lambda)$ be continuous for $\lambda \geqq 0$, $\varphi(0) = 0$, $\varphi \in C^\infty(0, \infty)$, and $d\varphi/d\lambda$ is completely monotone for $\lambda > 0$. Then $G(\lambda) = g(\varphi(\lambda))$ is completely monotone for $\lambda \geqq 0$.*

Substituting (5.5) into (5.1) gives the recursion formula

$$x_n(\lambda) = \sum_{j=1}^{n} x_{n-j}(\lambda) q_j / P\big(n + \nu(\lambda), \lambda\big) \quad \text{and } x_0(\lambda) = 1 . \tag{5.6}$$

Here $P(\nu, \lambda)$ is the indicial polynomial (5.3) for fixed λ, so that, by (5.4),

$$P\big(n + \nu(\lambda), \lambda\big) = 2n[(1 - r_0)^2/4 + q_0(\lambda)]^{\frac{1}{2}} + n^2 . \tag{5.7}$$

An analogue of Proposition 5.1 with $g(\mu) = (2n\mu + n^2)^{-1}$ and $\varphi(\lambda) = [(1 - r_0)^2/4 + q_0(\lambda)]^{\frac{1}{2}}$ shows that $G(\lambda) = g\big(\varphi(\lambda)\big)$ is completely monotone. Another application gives that $q_j/P\big(n + \nu(\lambda), \lambda\big)$ is completely monotone on Λ. An induction and (5.6) imply that $x_0(\lambda) = 1$, $x_1(\lambda)$, ... are completely monotone.

6. – A property of the Γ-function.

The modified Bessel differential equation (1.1) is of the type (5.1) with $q_0(\lambda) = \lambda = \mu^2 > 0$. The solution $I_\mu(t)$ has an expansion,

$$I_\mu(t) = [(t/2)^\mu/\Gamma(1 + \mu)] \cdot \Big[1 + \sum_{n=1}^{\infty} (t/2)^n / n!(\mu + 1) \dots (\mu + n)\Big] , \tag{6.1}$$

analogous to (5.6); cf. [15], p. 77. From Theorem 5.1, we can only deduce that $I_\mu(t)$ is a completely monotone function of $\lambda = \mu^2 > 0$ for $0 < t \leqq 2e^{-\gamma}$ (rather than for all $t > 0$). In fact, we have the following:

PROPOSITION 6.1. *Let $t > 0$ be fixed and $f(\mu) = t^\mu/\Gamma(1 + \mu)$. Then $F(\lambda) = f(\lambda^{\frac{1}{2}})$ is completely monotone for $\lambda \geqq 0$ if and only if*

$$0 < t \leqq e^{-\gamma} = 0.56 \dots , \tag{6.2}$$

where γ is the Euler-Mascheroni constant. In this case, $-\partial \log F(\lambda)/\partial\lambda$ is also completely monotone for $\lambda > 0$.

REMARK 1. By the Hausdorff-Bernstein theorem (cf. [16], p. 60), it follows that if (6.2) holds, then there exists a distribution function $V(r) = V(r, t)$ for $r \geqq 0$ satisfying

$$t^\mu/\Gamma(1 + \mu) = \int_0^\infty \exp(-r\mu^2)\, V(dr, t) \qquad \text{for } \mu > 0 ;$$

$V(r, t) = 0$ or $V(r, t) = V\big(r + \log(te^\gamma), e^{-\gamma}\big)$ according as $r + \log(te^\gamma) \leqq 0$ or > 0.

REMARK 2. The exponent $\frac{1}{2}$ in $F(\lambda) = f(\lambda^{\frac{1}{2}})$ is the « best » possible in the sense that $f(\lambda^{\delta})$ is not completely monotone on $\lambda > 0$, for any $\delta > \frac{1}{2}$ and any $t > 0$. In fact, it is readily verified, by using the formulas in the proof to follow, that $d^2 f(\lambda^{\delta})/d\lambda^2 \to -\infty$ as $\lambda \to 0$ if $\frac{1}{2} < \delta < 1$, $t > 0$.

PROOF. Let $\psi(\mu)$ be the logarithmic derivative of $\Gamma(\mu)$, $\psi(\mu) = \Gamma'(\mu)/\Gamma(\mu)$. By a standard formula,

$$\psi(1+\mu) + \gamma = \sum_{n=0}^{\infty} \mu/(n+1)(n+1+\mu) \geqq 0 \quad \text{for } \mu \geqq 0\,; \tag{6.3}$$

cf. [4], p. 15. Put

$$\varrho(\mu) = \int_0^{\mu} [\psi(1+s) + \gamma]\, ds \quad \text{and } \varphi(\lambda) = \varrho(\lambda^{\frac{1}{2}})\,. \tag{6.4}$$

Then Proposition 5.1 is applicable to $\varphi(\lambda)$ for, in this case,

$$d\varphi/d\lambda = [\psi(1+\lambda^{\frac{1}{2}}) + \gamma]/2\lambda^{\frac{1}{2}} = \sum_{n=0}^{\infty} 1/2(n+1)(n+1+\lambda^{\frac{1}{2}}) \tag{6.5}$$

is completely monotone for $\lambda \geqq 0$. By (6.4) and $f(\mu) = t^{\mu}/\Gamma(1+\mu)$,

$$F(\lambda) = f(\mu) = \exp\big(-\varrho(\mu)\big)\big((te^{\gamma})^{\mu} \quad \text{with } \mu = \lambda^{\frac{1}{2}} \geqq 0\,. \tag{6.6}$$

When (6.2) holds, $(te^{\gamma})^{\mu}$ is a completely monotone function of $\mu \geqq 0$, hence of $\lambda \geqq 0$ (by Proposition 5.1 with $\varphi(\lambda) = \lambda^{\frac{1}{2}}$). Also, $e^{-\varrho}$ is a completely monotone functjon of $\varrho \geqq 0$, so that $\exp\big(-\varrho(\mu)\big)$ is a completely monotone function of λ (by the case (6.4) of Proposition 5.1). Consequently, (6.6) is completely monotone for $\lambda \geqq 0$ when (6.2) holds. Also, in this case, $-F_{\lambda}(\lambda)/F(\lambda) = d\varphi/d\lambda + [\log(1/(te^{\gamma})]/2\lambda^{\frac{1}{2}}$ is completely monotone for $\lambda > 0$, where $d\varphi/d\lambda$ is given by (6.5).

Let $te^{\gamma} > 1$. Then, by (6.6), $df(\mu)/d\mu = [-d\varrho(\mu)/d\mu + \log(te^{\gamma})]f(\mu)$. Since $d\varrho/d\mu = 0$ at $\mu = 0$, by (6.3)-(6.4), and $\log(te^{\gamma}) > 0$, we have $df/du > 0$ at $\mu = 0$. Hence $F(\lambda) = f(\lambda^{\frac{1}{2}})$ is increasing for small $\lambda > 0$ and cannot be completely monotone.

Appendix 1: Monotone families of solutions.

We use the notation of Sections 1-3.

DEFINITION. *The class* $M_{\mu}(\Lambda)$. A function $h(\lambda)$, $\lambda \in \Lambda$, is said to be of class $M_1 = M_1(\Lambda)$ if h is nonnegative and nonincreasing. A function $h \in M_1(\Lambda)$

is said to be of class $M_2 = M_2(\Lambda)$ if it is continuous and convex on Λ. If $\mu = 2, 3, \ldots, h$ is said to be of class $M_\mu = M_\mu(\Lambda)$ if $h \in C^0(\Lambda) \cap C^{\mu-2}(\Lambda^0)$ and $(-1)^m h^{(m)} \in M_2(\Lambda^0)$ for $m = 0, \ldots, \mu - 2$. $M_\infty(\Lambda)$ is the class of completely monotone functions on Λ.

Note that $h \in M_\mu$ is continuous if $\mu > 1$. The class M_μ is closed under point-wise convergence if $\Lambda = \Lambda^0$ is open. This is false if if $\Lambda = [0, \infty)$ as is seen from the example $h_n(\lambda) = \exp(-n\lambda)$.

LEMMA. *Let $\Lambda = \Lambda^0$ be an open interval. Let $h_n \in M_\mu$ for $n = 1, 2, \ldots$ such that $h(\lambda) = \lim h_n(\lambda)$ exists as $n \to \infty$ for $\lambda \in \Lambda$. Then $h \in M_\mu$. Furthermore, if $\mu \geqq 2$, then*

$$h_n^{(j)}(\lambda) \to h^{(j)}(\lambda) \quad \text{as } n \to \infty \text{ for } j = 0, \ldots, \mu - 2 \tag{1}$$

uniformly on compacts of Λ.

PROOF. It is clear that $h \in M_\mu$ if $\mu = 1$ or $\mu = 2$. Let $\mu = 2$, $a < c < \beta$, and $[a, \beta) \subset \Lambda$. Then $c \leqq \sigma < \tau < \beta$ implies that

$$0 \leqq [h_n(\sigma) - h_n(\tau)]/(\tau - \sigma) \leqq [h_n(a) - h_n(c)]/(c - a),$$

so that h_n is uniformly Lipschitz continuous on $[c, \beta)$ with a Lipschitz constant independent of n. Thus (1) with $j = 0$ holds uniformly on λ-compacts.

Let $\mu = 3$. The argument just completed shows that the sequence of first order derivatives $h_1', h_2', \ldots$ are uniformly bounded on λ-compacts. Thus, since they are convex, they are uniformly Lipschitz continuous with a Lipschitz constant independent of n on λ-compacts. Hence the Arzela selection theorem implies that there exist subsequences of $h_1', h_2', \ldots$ uniformly convergent on λ-compacts. But the limit of such a subsequence is necessarily h', independent of the subsequence. Consequently, (2) with $j = 1$ holds uniformly on λ-compacts. This proves the case $\mu = 3$. The proof of the Lemma can be completed by a simple induction.

($\mathrm{A2}_{k\mu}$) *Let $\mu \geqq 1$ and $k > 0$ be fixed integers. Let $\partial^m q/\partial\lambda^m$ exist, be continuous, and satisfy $(-1)^{m+k}\partial^m q/\partial\lambda^m \geqq 0$ on $T \times \Lambda^0$ for $m = 1, 2, \ldots, \mu$ (in particular, $(-1)^{k+1}\partial q(t, \cdot)/\partial\lambda \in M_{\mu-1}(\Lambda^0)$).*

The Lemma just proved and the arguments in Section 2 and 3 have the following consequences which are analogues of statements in Sections 1-3.

THEOREM $1.1_{k\mu}$. *Assume* (A1) *and* ($\mathrm{A2}_{k\mu}$) *for fixed integers (k, μ), $0 < k < n$* and *$\mu > 1$. Let $\tau \in T^0$. Then $D^i \eta_{n-k}(t, \tau, \lambda) \in C^0(T \times T^0 \times \Lambda^0)$ for $0 \leqq i \leqq n$ and $\eta_{n-k}(t, \tau, \cdot) \in M_\mu(\Lambda^0)$ for fixed (t, τ), $\alpha < \tau \leqq t < \beta$.*

THEOREM 1.2_μ. *Assume* (A1), $(A2_{k\mu})$ *with* $k=n-1$ *and* $\mu>1$, *and* $(A3_\sigma)$ *with* $\sigma=\alpha$ [*or* $\sigma=\beta$]. *Then* $D^i\xi_1(t,\lambda)\in C^0(T\times\Lambda^0)$ *for* $0\leqq i\leqq n$, *and* $\xi_1(t,\cdot)$ [*or* $1/\xi_1(t,\cdot)$] *is of class* $M_\mu(\Lambda^0)$ *for fixed* $t\in T^0$.

LEMMA $2.1_{k\mu}$. *Assume* (A1), $(A2_{k\mu})$ *with fixed* (k,μ), $0<k<n$ *and* $\mu>0$. *Let* $[a,b]\subset T$. *Then* $(-1)^kG_{ab}(t,s,\cdot)\in M_\mu(\Lambda)$.

PROPOSITION $3.1_{k\mu}$. *Assume* (A1), $(A2_{k\mu})$ *with fixed* (k,μ), $0<k<n$ *and* $\mu>0$. *Then* $(-1)^{k+1}X^k(t,a,s,\cdot)\in M_\mu(\Lambda)$ *for fixed* (t,a,s), $\alpha<a<t<s<\beta$, *and* $X^{n-k}(t,b,s,\cdot)\in M_\mu(\Lambda)$ *for fixed* (t,b,s), $\alpha<s<t<b$.

Note that Λ (rather than Λ^0) and $\mu>0$ (rathe- than $\mu>1$) occur in the last two assertions, since no limit process is involved in the proof of Lemma $2.1_{k\mu}$ and the limit process in the proof of Proposition $3.1_{k\mu}$ is uniform on compacts of Λ.

Appendix 2: Cauchy functions of n-th order equations.

In the N-th order linear differential equation

$$Lx\equiv\left\{p_{m+1}^{-1}Dp_m^{-1}D\dots p_1^{-1}D^k-\sum_{j=0}^{k-1}q_jD^j\right\}x=0\,, \tag{1}$$

let $D=d/dt$; $k\geqq 1$; $m\geqq 0$; $n=k+m$; $p_j=p_j(t,\lambda)>0$, $q_j(t,\lambda)\geqq 0$ continuous on $T\times\Lambda$ such that $p_j(t,\cdot)$, $q_j(t,\cdot)$ are completely monotone on Λ (for fixed $t\in T$). For suitable functions $x(t)$ on T, define the vector $y(t;x)=(y_1,\dots,y_n)$ by $y_j=D^{j-1}x$ for $1\leqq j\leqq k$ and $y_{k+j}=p_1^{-1}D\dots Dp_j^{-1}D^kx$ for $1\leqq j\leqq m$, so that (1) is equivalent to the first order system

$$\begin{cases} y_j' &= y_{j+1} \qquad \text{for } 1\leqq j<k\,;\\ y_{k+j}' &= p_{j+1}y_{k+j+1} \qquad \text{for } 0\leqq j<m\,;\\ y_n' &= p_{m+1}\sum_{j=0}^{k-1}q_jy_{j+1}\,; \end{cases} \tag{2}$$

cf., e.g., [5], pp. 309-310.

For fixed λ, let $X(t,s,\lambda)$ be the Cauchy function for (1), i.e., if $y(t;x)$ is the vector belonging to the solution $x(t)=X(t,s,\lambda)$ of (1), then y satisfies the initial conditions

$$y_j=0 \quad \text{for } 1\leqq j<n \qquad \text{and} \qquad y_n=1 \quad \text{at } t=s\,. \tag{3}$$

We denote this vector $y(t; x)$, with $x = X(t, s, \lambda)$, by $Y(t, s, \lambda) = (Y_1(t, s, \lambda), \ldots, Y_n(t, s, \lambda))$. For example, $Y_j(t, s, \lambda) = D^{j-1}X(t, s, \lambda)$ for $1 \leqq j \leqq k$.

THEOREM. *Under the conditions above, $Y_j(t, s, \cdot)$ is completely monotone on Λ for fixed $t, s \in T$, $t > s$, and $j = 1, \ldots, n$.*

Note that, in this theorem, $q_j(t, \cdot)$ is completely monotone, so that assumption $(A2_0)$ holds, but the disconjugacy of (1) is not assumed. It is clear that the theorem is contained in the following

LEMMA. *Let $A(t, \lambda)$ be a continuous $n \times n$ matrix function on $T \times \Lambda$ such that every entry is completely monotone on Λ, for fixed $t \in T$. Let $U(t) = U(t, s, \lambda)$ be the fundamental matrix of*

$$U' = A(t, \lambda)\, U \tag{4}$$

reducing to the identity at $t = s$. Then each entry of $U(t, s, \cdot)$ is completely monotone on Λ for fixed $s, t \in T$, $t > s$.

REMARK. – It follows that if $h(t, \lambda)$ is a continuous vector on $T \times \Lambda$ such that each component of $h(t, \cdot)$ is completely monotone on Λ and $c(\lambda)$ is a vector with completely monotone components, then

$$u' = A(t, \lambda)u + h(t, \lambda), \qquad u(s) = c(\lambda),$$

has the solution

$$u(t, s, \lambda) = U(t, s, \lambda)c(\lambda) + \int_s^t U(t, r, \lambda)h(r, \lambda)\, dr,$$

with completely monotone components on Λ, for fixed $t, s \in T$, $t \geqq s$.

PROOF OF LEMMA. Since each entry of $A(t, \lambda)$ is non-negative, it is clear that each entry of $U(t, s, \lambda)$ is non-negative for $t \geqq s$. Let $U = U(t, s, \lambda)$ in (4) and differentiate with respect to $\lambda \in \Lambda^0$ to get

$$U'_\lambda(\cdot, s, \lambda) = A(\cdot, \lambda)\, U_\lambda(\cdot, s, \lambda) + A_\lambda(\cdot, \lambda)\, U(\cdot, s, \lambda).$$

Since $U_\lambda(t, s, \lambda) = 0$ when $t = s$, the variations of constants formula gives

$$U_\lambda(t, s, \lambda) = \int_s^t U(t, r, \lambda) A_\lambda(r, \lambda)\, U(r, s, \lambda)\, dr.$$

This shows that each entry of $U_\lambda(t, s, \lambda)$ is non-positive for $t \geqq s$. Repeated differentiations with respect to $\lambda \in \Lambda^0$ and an induction give the lemma.

Appendix 3: Special principal solutions.

Let the differential equation

$$D^n x + q_{n-1}(t) D^{n-1} x + \dots + q_1(t) Dx + q_0(t) x = 0 \tag{1}$$

have continuous coefficients on a t-interval T with endpoints α, β, where $-\infty \leqq \alpha < \beta \leqq \infty$. Let $T^0 = \operatorname{int} T$.

THEOREM. *Let* (1) *be disconjugate on* T *and* $[s, b] \subset T_0$. *Let* $x = X^j(t, b, s)$ *be the solution satisfying the boundary conditions*

$$\begin{aligned} &[D^{i-1} X^j]_{t=b} = 0 \quad \text{if } 1 \leqq i \leqq n-j\,, \\ &[D^{i-1} X^j]_{t=s} = 0 \quad \text{or } 1 \text{ if } 1 \leqq i < j \text{ or } i = j\,. \end{aligned} \tag{2}$$

Then

$$\eta_j(t, s) = \lim_{b \to \beta} X^j(t, b, s) \qquad \text{for } j = 1, \dots, n-1 \tag{3}$$

exists in C^n *on arbitrary* (t, s)*-compacts of* $T \times T^0$ *and is the* j*-th special principal solution of* (1) *at* $t = \beta$, *determined by* s; *cf.* (1.10) *above.*

PROOF. The case $j = 1$ follows from Theorem 7.1_n(ii), [7], p. 330. In particular, the theorem is correct if $n = 2$. Assume its validity for disconjugate differential equations of order $n-1 (\geqq 1)$. Let $1 < j < n$.

Let $\eta_1 = \eta_1(t, s)$. There exists a disconjugate differential equation of order $n-1$, say $L_{n-1} v = 0$, such that x is a solution of (1) if and only if $v = W(\eta_1, x) \equiv \eta_1 x' - \eta_1' x$ is a solution of $L_{n-1} v = 0$; furthermore, for $1 < j < n$, x is a j-th principal solution of (1) at $t = \beta$ if and only if $v = W(\eta_1, x)$ is a $(j-1)$-st principal solution of $L_{n-1} v = 0$ at $t = \beta$; Theorem 7.2_n(iv), [7], p. 332. Put $V^{j-1}(t, b, s) = W(\eta_1, X^j(t, b, s))$ for $j = 2, \dots, n-1$. Then $v = V^{j-1}$ is a solution of $L_{n-1} v = 0$ and satisfies the same boundary conditions at $t = b, s$ as does $X^{j-1}(t, b, s)$, i.e., (2) with j replaced by $j-1$. By the induction hypothesis, $\psi_{j-1}(t, s) = \lim W(\eta_1, X^j(t, b, s))$, as $b \to \beta$, exists in C^{n-1} on arbitrary (t, s)-compacts of $T \times T^0$, and is the $(j-1)$-st special principal solution of $L_{n-1} v = 0$ at $t = \beta$, determined by s. This implies that $(X^j/\eta_1)' \to \psi_{j-1}/\eta_1^2$ as $b \to \beta$. Hence, also $\varrho(t) = \lim X^j(t, b, s)$ exists, as $b \to \beta$, in C^n on arbitrary (t, s)-compacts of $T \times T^0$, and

$$\varrho(t) = \eta_1(t, s) \int_s^t [\psi_{j-1}(r, s)/\eta_1^2(r, s)]\, dr + c\eta_1(t, s)\,,$$

where c is a constant. Since $\varrho(s) = 0$ and $\eta_1 > 0$, we have $c = 0$, so that ϱ satisfies the condition (1.10) with $\tau = s$. Also, from $\psi_{j-1} = W(\eta_1, \varrho)$, it follows that $x = \varrho(t)$ is a j-th principal solution of (1) at $t = \beta$. Hence $\varrho(t) = \eta_j(t, s)$, and the proof is complete.

REFERENCES

[1] E. S. Čičin, *A theorem on a differential inequality for multi-point boundary value problems*, Izv. Vysš. Učebn. Zaved. Matematika, **2** (27) (1962), pp. 170-179 (Russian).

[2] W. A. Coppel, *Disconjugacy*, Lecture Notes in Mathematics, No. 220, Springer, 1971.

[3] J. L. Doob, *Stochastic Processes*, Wiley, New York, 1953.

[4] A. Erdelyi (editor), *Higher Transcendental Functions*, vol. 1, McGraw-Hill, New York, 1953.

[5] P. Hartman, *Ordinary Differential Equations*, Baltimore, 1973.

[6] P. Hartman, *Unrestricted n-parameter families*, Rend. Circ. Mat. Palermo (2), **7** (1958), pp. 123-142.

[7] P. Hartman, *Principal solutions of disconjugate n-th order linear differential equations*, Amer. J. Math., **91** (1969), pp. 306-362; **93** (1971), pp. 439-451.

[8] P. Hartman, *On N-parameter families and interpolation problems for nonlinear ordinary differential equations*, Trans. Amer. Math. Soc., **154** (1971), pp. 201-226.

[9] P. Hartman - G. S. Watson, *« Normal » distribution functions on spheres and the modified Bessel functions*, Ann. Prob., **2** (1974), pp. 593-607.

[10] A. Yu. Levin, *Some problems bearing on the oscillation of solutions of linear differential equations*, Dokl. Akad. Nauk SSSR, **148** (1963), pp. 512-515; *Soviet Math. Dokl.*, **4** (1963), pp. 121-124.

[11] A. Yu. Levin, *Nonoscillation of solutions of the equation* $x^{(n)} + p_1(t)x^{(n-1)} + \ldots = 0$, Uspechi Mat. Nauk, **24** (1969), No. 2 (1946), pp. 43-96; Russian Math. Surveys, **24** (1969), pp. 43-99.

[12] H. P. McKean jr., *Elementary solutions for certain parabolic partial differential equations*, Trans. Amer. Math. Soc., **82** (1950), pp. 519-548.

[13] Z. Opial, *On a theorem of O. Arama*, J. Differential Equations, **3** (1967), pp. 88-91.

[14] Ju. V. Pokornyĭ, *Some estimates of the Green's functions of a multipoint boundary value problem*, Mat. Zametki, **4** (1968), pp. 533-540 (Russian).

[15] G. N. Watson, *A Treatise on the Theory of Bessel Functions*, 2nd ed., Cambridge, 1958.

[16] D. Widder, *The Laplace Transform*, Princeton, 1941.